

Polynomial Regression Assignment

Geothermal Power Plant Expansion Project

Roll No.: BT2024136

All code is available in the GitHub repository:

https://github.com/yashvi2706/BT2024136_ML_Assignment1.

1 Introduction

This report describes polynomial regression models for two personalised datasets. **var1** predicts a Net Power Score from six turbine parameters $(x_1, \dots, x_6)$, and **var2** predicts a Thermal Anomaly Score from three spatial coordinates (x_1, x_2, x_3) . Each training set has 1000 samples and each test set has 1000 samples. Only polynomial regression was used, as required.

2 Method

Both problems use the same pipeline:

1. Expand the inputs into all polynomial terms up to total degree d (the sum of the powers in each term is at most d), without a separate bias column.
2. Standardise every polynomial term to zero mean and unit variance. This keeps high-degree terms numerically stable and makes the regularisation penalty treat all terms comparably.
3. Fit a regularised linear model on the standardised terms.

The degree d and the regularisation strength α were chosen by 5-fold cross-validation (CV) on the training data, using mean squared error (MSE) as the criterion. Folds were shuffled with a fixed random seed.

The two regularisers were

$$\begin{aligned}\text{Ridge: } & \min_w \|y - \Phi w\|_2^2 + \alpha \|w\|_2^2, \\ \text{Lasso: } & \min_w \frac{1}{2n} \|y - \Phi w\|_2^2 + \alpha \|w\|_1,\end{aligned}$$

where Φ is the standardised polynomial feature matrix. Ridge shrinks all coefficients and is well suited when most terms carry some signal. Lasso sets many coefficients exactly to zero and is better when the true function uses only a subset of terms.

3 Data Observations

In both datasets, roughly 25–30% of the values of each input equal exactly -1 or $+1$. This suggests the inputs were clipped to $[-1, 1]$. Clipping makes the observed relationship between inputs and target slightly different from a pure polynomial, which probably puts a floor under the achievable error. The target ranges are about $[-10.4, 12.4]$ for var1 and $[-29.3, 39.3]$ for var2.

4 Problem 1 (var1): Turbine Optimisation

The problem statement says the underlying polynomial has degree up to 10. With six inputs, a degree- d expansion has $\binom{d+6}{6} - 1$ terms (27, 83, 209, 461, ... for $d = 2, 3, 4, 5$), which is large compared with 1000 samples. Regularisation is therefore essential.

Degree selection. With ridge regression, CV error fell sharply from degree 1 to degree 4 and was lowest at degree 5; degrees 6 and 7 were worse (Figure 1). Unregularised least squares overfit beyond degree 4 (CV MSE about 0.80 at degree 4 and above 2 at degree 5).

Model choice. Lasso at degree 5 reached a lower CV error than ridge at the same degree, and the lasso solution keeps only 112 of 461 terms, so the underlying function appears sparse. Degrees 6 and 7 with lasso were slightly worse than degree 5. The final model is therefore **degree 5 with Lasso** ($\alpha = 0.01$).

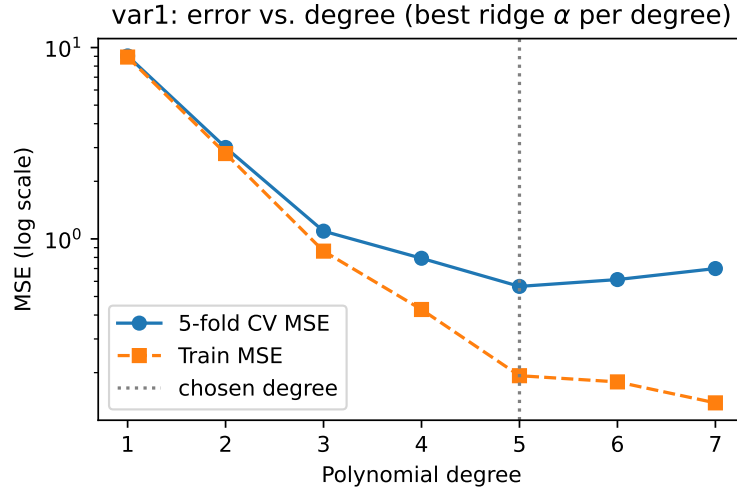


Figure 1: var1: CV and training MSE against polynomial degree (ridge, best α at each degree). Lasso at degree 5 achieves a lower CV MSE than the ridge value plotted there.

5 Problem 2 (var2): Thermal Reservoir Mapping

The problem statement says the underlying polynomial has degree up to 20. With three inputs, degree 10 has 285 terms and degree 14 has 679, so even high degrees are manageable for 1000 samples.

Degree selection. CV error dropped steadily up to degree 8 and then plateaued near 0.23–0.25 for degrees 9 to 14 (Figure 2), while training error kept falling. Beyond degree 10 the extra terms reduce training error without improving CV error, so I chose the simplest model on the plateau.

Model choice. Ridge regression with $\alpha = 1$ at **degree 10** gave a CV MSE of about 0.24. Lasso variants did not beat ridge here (best about 0.25 after refitting selected terms), which is consistent with most terms contributing to the fit (Lasso at $\alpha = 0.003$ for degree 10 gave 0.28).

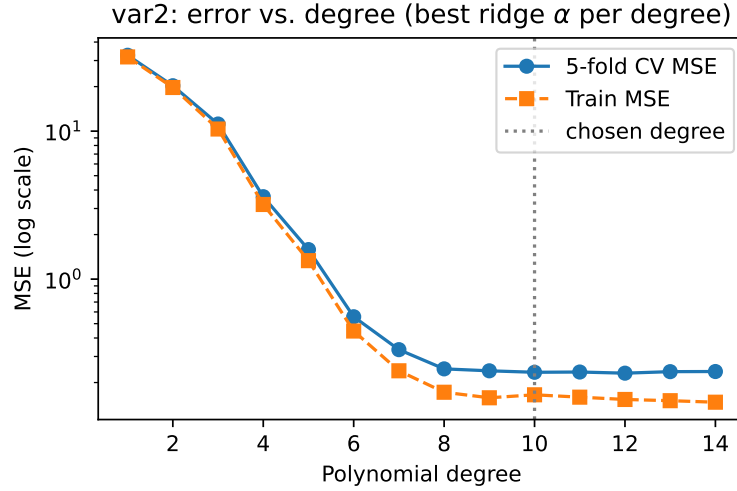


Figure 2: var2: CV and training MSE against polynomial degree (ridge, best α at each degree).

6 Results

Table 1 summarises the final models. The test targets are hidden, so the CV scores are the estimate of test performance; training scores are shown only to indicate the size of the train–CV gap.

Table 1: Final models. CV values are 5-fold averages over two different fold shuffles.

Problem	Model	Degree	α	CV MSE	CV R^2	Train MSE / R^2
var1	Lasso	5	0.01	0.35	0.963	0.26 / 0.973
var2	Ridge	10	1.0	0.24	0.994	0.17 / 0.996

In both problems the gap between training and CV error is small, which indicates the chosen models are not heavily overfitted.

7 Other Techniques and Things Tried

- Plain least squares at all degrees (shown to overfit at high degree).
- Ridge regression with a grid of α for every degree.
- Lasso with a grid of α , and a “relaxed lasso” variant (select terms with lasso, then refit them by least squares). The relaxed version was not better than plain lasso.
- Repeating cross-validation with a second random shuffle to confirm that the scores are stable (differences of about 0.01 in MSE).

8 Conclusion

For var1 a sparse degree-5 polynomial fitted with Lasso gives CV $R^2 \approx 0.963$. For var2 a degree-10 polynomial fitted with Ridge gives CV $R^2 \approx 0.994$. Predictions were written to `BT2024136_pred_var1.csv` and `BT2024136_pred_var2.csv`.